9. Simulation and Performance Evaluation of OFDMA Resource-Block Allocation for

Multi-User Communication

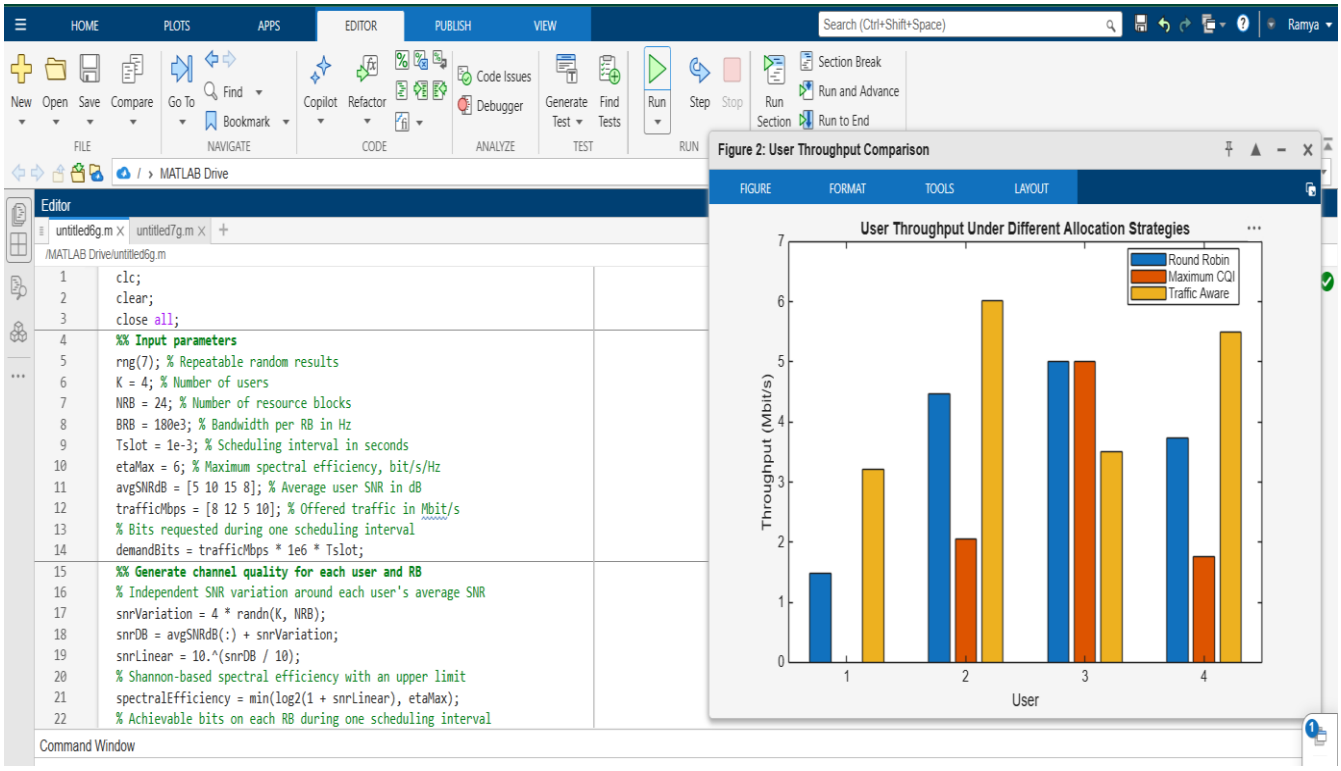
PROGRAM:

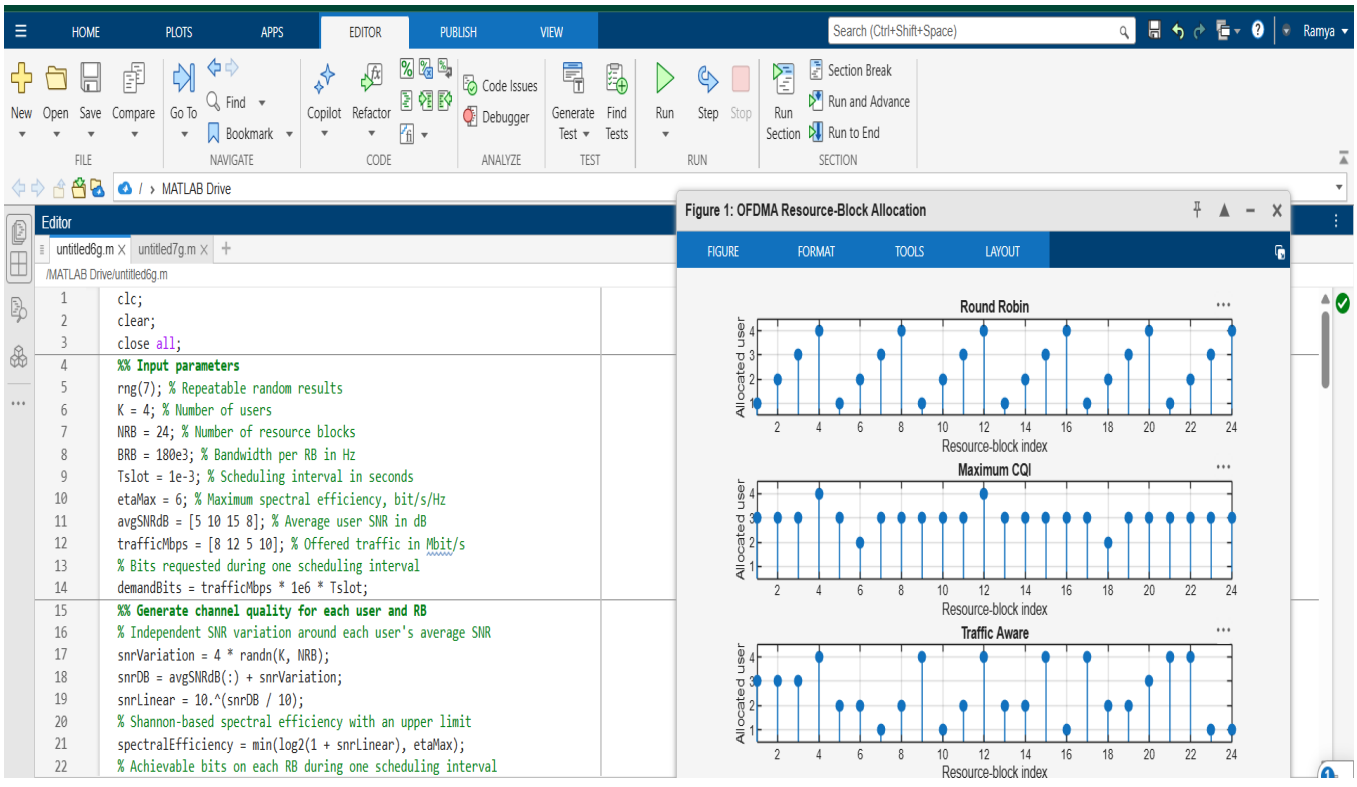
```
clc;
clear;
close all;
%% Input parameters
rng(7); % Repeatable random results
K = 4; % Number of users
NRB = 24; % Number of resource blocks
BRB = 180e3; % Bandwidth per RB in Hz
Tslot = 1e-3; % Scheduling interval in seconds
etaMax = 6; % Maximum spectral efficiency, bit/s/Hz
avgSNRdB = [5 10 15 8]; % Average user SNR in dB
trafficMbps = [8 12 5 10]; % Offered traffic in Mbit/s
% Bits requested during one scheduling interval
demandBits = trafficMbps * 1e6 * Tslot;
%% Generate channel quality for each user and RB
% Independent SNR variation around each user's average SNR
snrVariation = 4 * randn(K, NRB);
snrDB = avgSNRdB(:) + snrVariation;
snrLinear = 10.^(snrDB / 10);
% Shannon-based spectral efficiency with an upper limit
spectralEfficiency = min(log2(1 + snrLinear), etaMax);
% Achievable bits on each RB during one scheduling interval
rateBps = BRB * spectralEfficiency;
bitsPerRB = rateBps * Tslot;
strategyNames = {'Round Robin', 'Maximum CQI', 'Traffic Aware'};
```

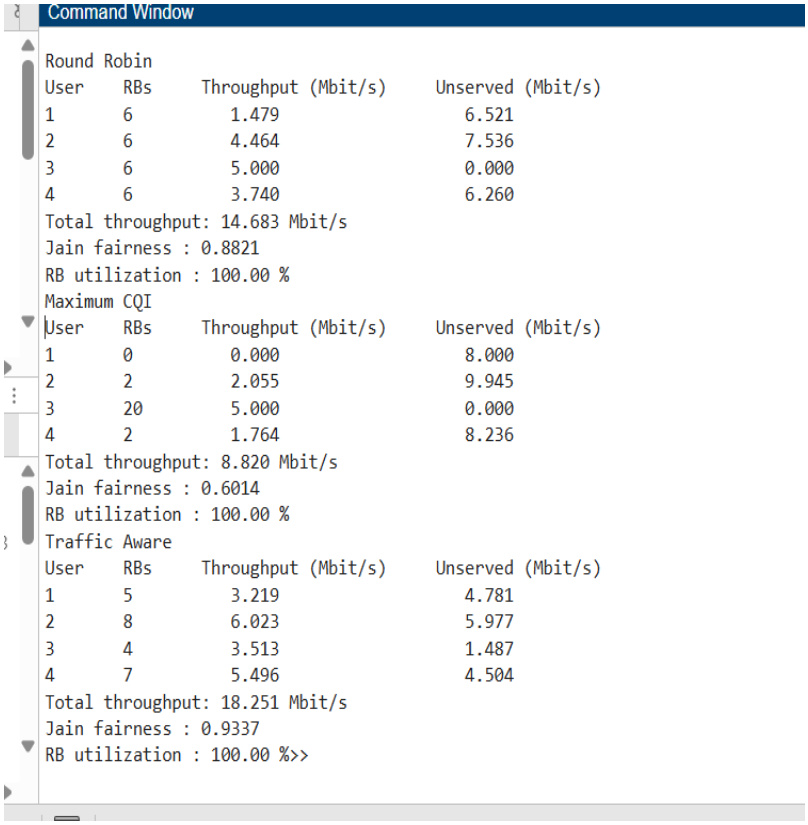
```
allocations = zeros(K, NRB, 3);
%% Strategy 1: Round-robin allocation
for n = 1:NRB
user = mod(n - 1, K) + 1;
allocations(user, n, 1) = 1;
end
%% Strategy 2: Maximum-CQI allocation
for n = 1:NRB
[~, user] = max(snrDB(:, n));
allocations(user, n, 2) = 1;
end
%% Strategy 3: Traffic-aware channel-quality allocation
servedBits = zeros(K, 1);
for n = 1:NRB
remainingBits = max(demandBits(:) - servedBits, 0);
if all(remainingBits <= 0)
break;
end
normalizedDemand = remainingBits ./ max(demandBits(:), eps);
metric = bitsPerRB(:, n) .* normalizedDemand;
% Prevent allocation to a user whose demand has been satisfied
metric(remainingBits <= 0) = -inf;
[~, user] = max(metric);
allocations(user, n, 3) = 1;
deliveredBits = min(bitsPerRB(user, n), remainingBits(user));
servedBits(user) = servedBits(user) + deliveredBits;
end
%% Performance calculations
userThroughputMbps = zeros(K, 3);
```

```
totalThroughputMbps = zeros(1, 3);
jainFairness = zeros(1, 3);
utilization = zeros(1, 3);
unservedTrafficMbps = zeros(K, 3);
rbPerUser = zeros(K, 3);
for s = 1:3
allocation = allocations(:, :, s);
deliveredBits = sum(allocation .* bitsPerRB, 2);
% Throughput is restricted to requested traffic
usefulBits = min(deliveredBits, demandBits(:));
userThroughputMbps(:, s) = usefulBits / Tslot / 1e6;
totalThroughputMbps(s) = sum(userThroughputMbps(:, s));
throughput = userThroughputMbps(:, s);
jainFairness(s) = sum(throughput)^2 / (K * sum(throughput.^2) + eps);
allocatedRBs = sum(allocation(:));
utilization(s) = allocatedRBs / NRB * 100;
unservedTrafficMbps(:, s) = max(trafficMbps(:) - throughput, 0);
rbPerUser(:, s) = sum(allocation, 2);
end
%% Display results
fprintf('\nOFDMA RESOURCE-BLOCK ALLOCATION RESULTS\n');
fprintf('=======================================\n');
for s = 1:3
fprintf('\n%s\n', strategyNames{s});
fprintf('User\tRBs\tThroughput (Mbit/s)\tUnserved (Mbit/s)\n');
for k = 1:K
fprintf('%d\t%d\t%8.3f\t\t%8.3f\n', ...
k, rbPerUser(k, s), userThroughputMbps(k, s), unservedTrafficMbps(k, s));
end
```

```
fprintf('Total throughput: %.3f Mbit/s\n', totalThroughputMbps(s));
fprintf('Jain fairness : %.4f\n', jainFairness(s));
fprintf('RB utilization : %.2f %%%\n', utilization(s));
end
%% Allocation maps plotting
figure('Name', 'OFDMA Resource-Block Allocation');
for s = 1:3
subplot(3, 1, s);
[userIndex, rbIndex] = find(allocations(:, :, s));
stem(rbIndex, userIndex, 'filled', 'LineWidth', 1.1);
grid on;
xlim([1 NRB]); ylim([0.5 K + 0.5]); yticks(1:K);
xlabel('Resource-block index'); ylabel('Allocated user');
title(strategyNames{s});
end
%% Throughput comparison plotting
figure('Name', 'User Throughput Comparison');
bar(userThroughputMbps);
xlabel('User'); ylabel('Throughput (Mbit/s)');
title('User Throughput Under Different Allocation Strategies');
legend(strategyNames, 'Location', 'best');
```


HOME
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EDITOR
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Search (Ctrl+Shift+Space)
Ramya
New
Open
Save
Compare
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Find
Bookmark
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Code Issues
Debugger
Generate Test
Find Tests
Run
Step
Stop
Run Section
Section Break
Run and Advance
Run to End
FILE
NAVIGATE
CODE
ANALYZE
TEST
RUN
/ › MATLAB Drive
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untitled6g.m
untitled7g.m
/MATLAB Drive/untitled6g.m
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2 clear;
3 close all;
4 %% Input parameters
5 rng(7); % Repeatable random results
6 K = 4; % Number of users
7 NRB = 24; % Number of resource blocks
8 BRB = 180e3; % Bandwidth per RB in Hz
9 Tslot = 1e-3; % Scheduling interval in seconds
10 etaMax = 6; % Maximum spectral efficiency, bit/s/Hz
11 avgSNRdB = [5 10 15 8]; % Average user SNR in dB
12 trafficMbps = [8 12 5 10]; % Offered traffic in Mbit/s
13 % Bits requested during one scheduling interval
14 demandBits = trafficMbps * 1e6 * Tslot;
15 %% Generate channel quality for each user and RB
16 % Independent SNR variation around each user's average SNR
17 snrVariation = 4 * randn(K, NRB);
18 snrDB = avgSNRdB(:) + snrVariation;
19 snrLinear = 10.^(snrDB / 10);
20 % Shannon-based spectral efficiency with an upper limit
21 spectralEfficiency = min(log2(1 + snrLinear), etaMax);
22 % Achievable bits on each RB during one scheduling interval
Command Window
Figure 2: User Throughput Comparison
FIGURE
FORMAT
TOOLS
LAYOUT
User Throughput Under Different Allocation Strategies
Round Robin
Maximum CQI
Traffic Aware
Throughput (Mbit/s)
User


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Ramya
New
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Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run
Step
Stop
Run Section
Section Break
Run and Advance
Run to End
FILE
NAVIGATE
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RUN
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Figure 1: OFDMA Resource-Block Allocation
FIGURE
FORMAT
TOOLS
LAYOUT
Round Robin
Maximum CQI
Traffic Aware
Allocated user
Resource-block index


Command Window
Round Robin
User RBs Throughput (Mbit/s) Unserved (Mbit/s)
1 6 1.479 6.521
2 6 4.464 7.536
3 6 5.000 0.000
4 6 3.740 6.260
Total throughput: 14.683 Mbit/s
Jain fairness : 0.8821
RB utilization : 100.00 %
Maximum CQI
User RBs Throughput (Mbit/s) Unserved (Mbit/s)
1 0 0.000 8.000
2 2 2.055 9.945
3 20 5.000 0.000
4 2 1.764 8.236
Total throughput: 8.820 Mbit/s
Jain fairness : 0.6014
RB utilization : 100.00 %
Traffic Aware
User RBs Throughput (Mbit/s) Unserved (Mbit/s)
1 5 3.219 4.781
2 8 6.023 5.977
3 4 3.513 1.487
4 7 5.496 4.504
Total throughput: 18.251 Mbit/s
Jain fairness : 0.9337
RB utilization : 100.00 %>>